Отчет по лабораторной работе № 6

Сетевые технологии

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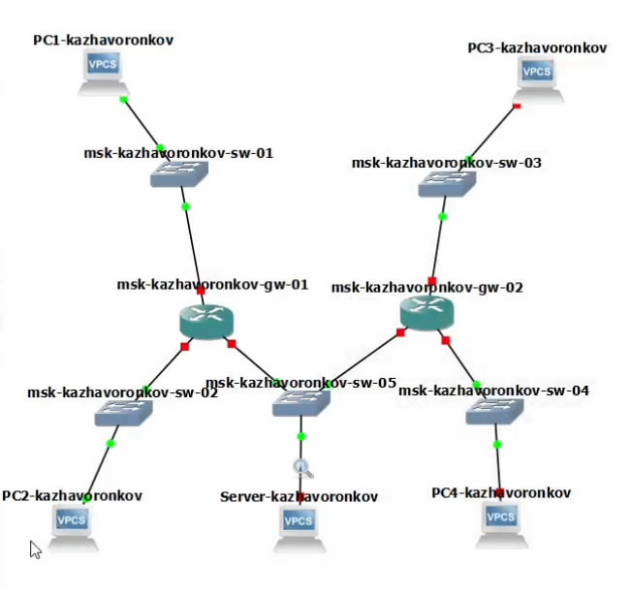
Содержание i

1. Вывод

0.1 Цель

Изучение принципов распределения и настройки адресного пространства на устройствах сети.

0.2 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети



0.3 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

Включим захват трафика.

0.4 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

Настроим IPv4-адресацию для интерфейсов узлов PC1, PC2, Server:

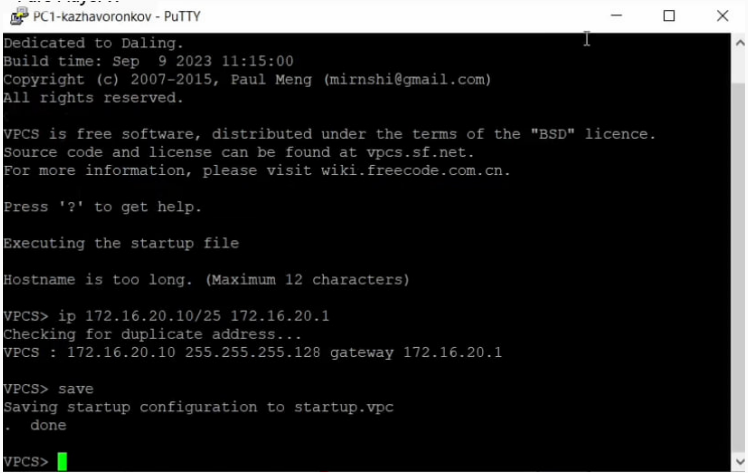


Рисунок 3: Настройка PC1

0.5 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

PC2:

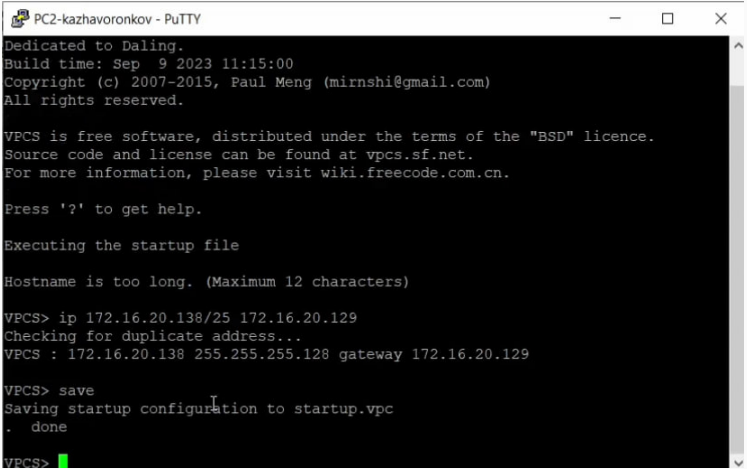


Рисунок 4: Настройка PC2

0.6 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

Server:

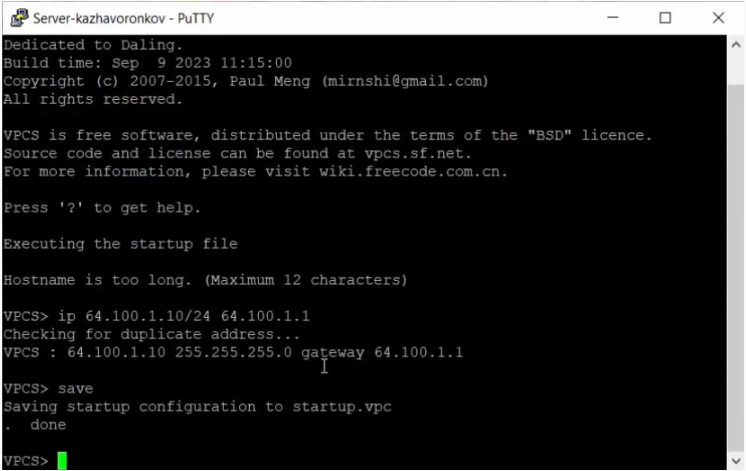


Рисунок 5: Настройка Server

0.7 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

PC1-kazhavoronkov - PuTTY

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 172.16.20.10/25
GATEWAY     : 172.16.20.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 20022
RHOST:PORT  : 127.0.0.1:20023
MTU         : 1500

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE       :
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:02
LPORT              : 20022
RHOST:PORT         : 127.0.0.1:20023
MTU:               : 1500

VPCS>
```

Рисунок 6: Просмотр конфигураций

0.8 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

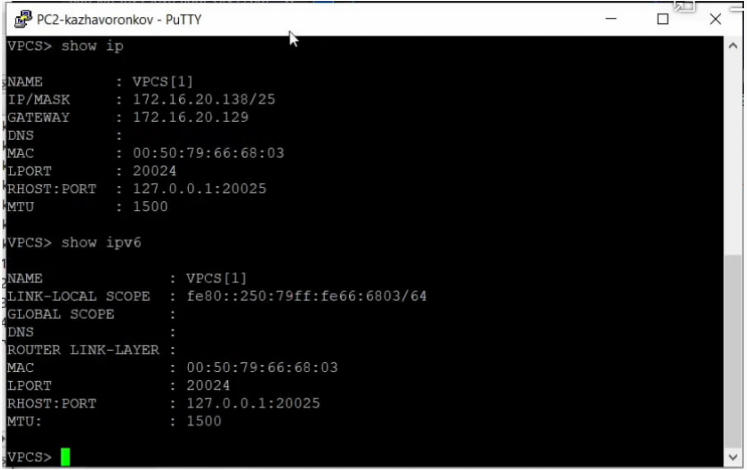


Рисунок 7: Просмотр конфигураций

0.9 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-01 - PuTTY

```
frr(config)# hastname msk-kazhavoronkov-gw-01
% Unknown command: hastname msk-kazhavoronkov-gw-01
frr(config)# hostname msk-kazhavoronkov-gw-01
msk-kazhavoronkov-gw-01(config)# exit
msk-kazhavoronkov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-kazhavoronkov-gw-01# configure terminal
msk-kazhavoronkov-gw-01(config)# interface eth0
msk-kazhavoronkov-gw-01(config-if)# ip address 172.16.20.1/25
msk-kazhavoronkov-gw-01(config-if)# no shutdown
msk-kazhavoronkov-gw-01(config-if)# exit
msk-kazhavoronkov-gw-01(config)# interface eth1
msk-kazhavoronkov-gw-01(config-if)# ip address 172.16.20.129/25
msk-kazhavoronkov-gw-01(config-if)# no shutdown
msk-kazhavoronkov-gw-01(config-if)# exit
msk-kazhavoronkov-gw-01(config)# interface eth2
msk-kazhavoronkov-gw-01(config-if)# ip address 64.100.1.1/24
msk-kazhavoronkov-gw-01(config-if)# no shutdown
msk-kazhavoronkov-gw-01(config-if)# exit
msk-kazhavoronkov-gw-01(config)# exit
msk-kazhavoronkov-gw-01# write memo
```

Рисунок 8: Настройка FRR

0.10 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-01 - PuTTY

```
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-kazhavoronkov-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 172.16.20.1/25
exit
!
interface eth1
 ip address 172.16.20.129/25
exit
!
interface eth2
 ip address 64.100.1.1/24
exit
!
end
msk-kazhavoronkov-gw-01# show interface brief
Interface       Status  VRF             Addresses
---------       ------  ---             ---------
eth0            up      default         172.16.20.1/25
eth1            up      default         172.16.20.129/25
eth2            up      default         64.100.1.1/24
eth3            down    default
eth4            down    default
eth5            down    default
eth6            down    default
eth7            down    default
lo              up      default
pimreg          up      default

msk-kazhavoronkov-gw-01#
```

0.11 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-01 - PuTTY

```
eth2            up      default         64.100.1.1/24
eth3            down    default
eth4            down    default
eth5            down    default
eth6            down    default
eth7            down    default
lo              up      default
pimreg          up      default

msk-kazhavoronkov-gw-01# ping 172.16.20.1/25
ping: bad address '172.16.20.1/25'
msk-kazhavoronkov-gw-01# ping 172.16.20.1
PING 172.16.20.1 (172.16.20.1): 56 data bytes
64 bytes from 172.16.20.1: seq=0 ttl=64 time=0.923 ms
64 bytes from 172.16.20.1: seq=1 ttl=64 time=0.054 ms
64 bytes from 172.16.20.1: seq=2 ttl=64 time=0.045 ms
^C
--- 172.16.20.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.045/0.340/0.923 ms
msk-kazhavoronkov-gw-01# ping -c 3 172.16.20.129
% Unknown command: ping -c 3 172.16.20.129
msk-kazhavoronkov-gw-01# ping  172.16.20.129
PING 172.16.20.129 (172.16.20.129): 56 data bytes
64 bytes from 172.16.20.129: seq=0 ttl=64 time=0.179 ms
64 bytes from 172.16.20.129: seq=1 ttl=64 time=0.756 ms
64 bytes from 172.16.20.129: seq=2 ttl=64 time=0.137 ms
^C
--- 172.16.20.129 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.137/0.357/0.756 ms
msk-kazhavoronkov-gw-01# trace 172.16.20.1
traceroute to 172.16.20.1 (172.16.20.1), 30 hops max, 46 byte packets
 1
^Cmsk-kazhavoronkov-gw-01#
```

0.12 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-01 - PuTTY

```
eth2            up      default         64.100.1.1/24
eth3            down    default
eth4            down    default
eth5            down    default
eth6            down    default
eth7            down    default
lo              up      default
pimreg          up      default

msk-kazhavoronkov-gw-01# ping 172.16.20.1/25
ping: bad address '172.16.20.1/25'
msk-kazhavoronkov-gw-01# ping 172.16.20.1
PING 172.16.20.1 (172.16.20.1): 56 data bytes
64 bytes from 172.16.20.1: seq=0 ttl=64 time=0.923 ms
64 bytes from 172.16.20.1: seq=1 ttl=64 time=0.054 ms
64 bytes from 172.16.20.1: seq=2 ttl=64 time=0.045 ms
^C
--- 172.16.20.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.045/0.340/0.923 ms
msk-kazhavoronkov-gw-01# ping -c 3 172.16.20.129
% Unknown command: ping -c 3 172.16.20.129
msk-kazhavoronkov-gw-01# ping  172.16.20.129
PING 172.16.20.129 (172.16.20.129): 56 data bytes
64 bytes from 172.16.20.129: seq=0 ttl=64 time=0.179 ms
64 bytes from 172.16.20.129: seq=1 ttl=64 time=0.756 ms
64 bytes from 172.16.20.129: seq=2 ttl=64 time=0.137 ms
^C
--- 172.16.20.129 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.137/0.357/0.756 ms
msk-kazhavoronkov-gw-01# trace 172.16.20.1
traceroute to 172.16.20.1 (172.16.20.1), 30 hops max, 46 byte packets
 1
^Cmsk-kazhavoronkov-gw-01#
```

0.13 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

Настроим IPv6-адресацию для интерфейсов узлов PC3, PC4, Server

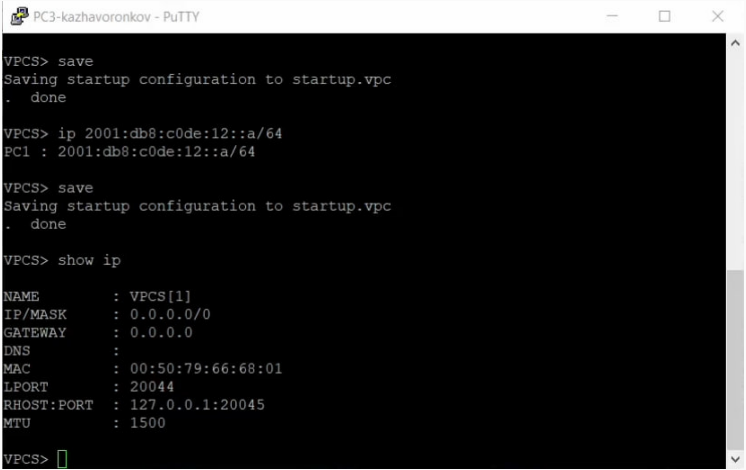


Рисунок 12: Настройка PC3

PC4:

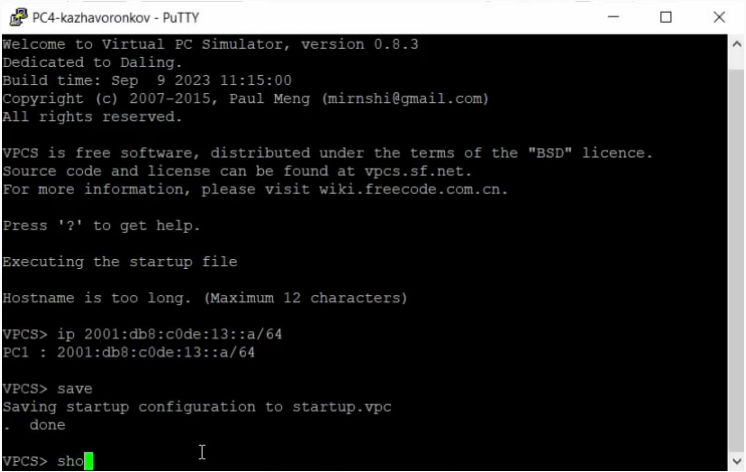


Рисунок 13: Настройка PC4

Server:

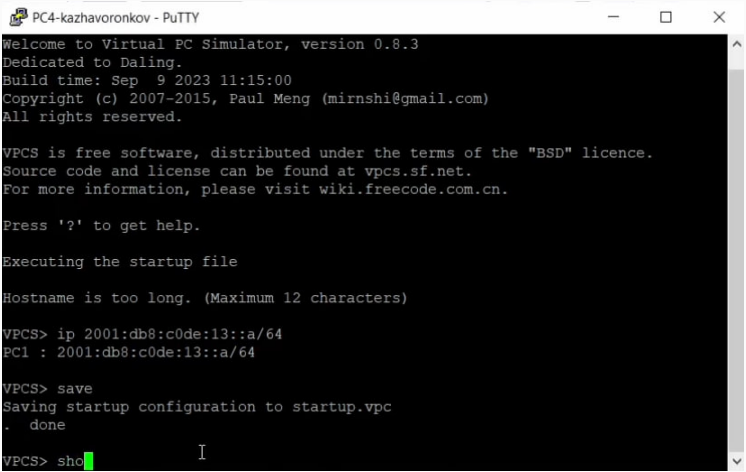


Рисунок 14: Настройка Server

0.16 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

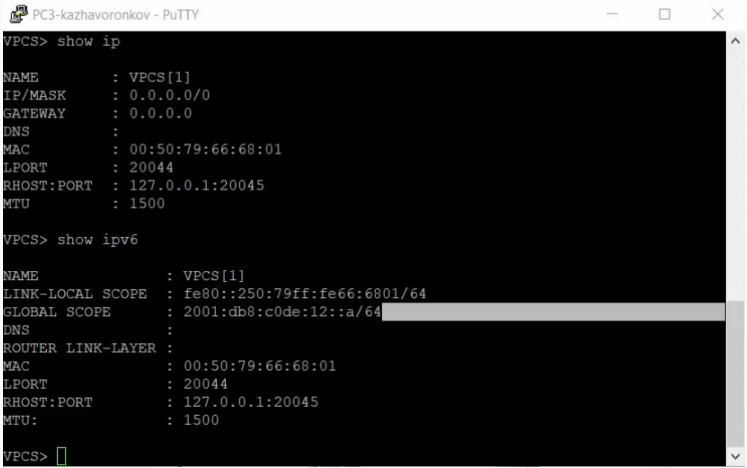


Рисунок 15: Просмотр конфигураций

0.17 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

PC4-kazhavoronkov - PuTTY

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 0.0.0.0/0
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:04
LPORT       : 20046
RHOST:PORT  : 127.0.0.1:20047
MTU         : 1500

VPCS> show ipv6

NAME               : VPCS[1]
LINK-LOCAL SCOPE   : fe80::250:79ff:fe66:6804/64
GLOBAL SCOPE       : 2001:db8:c0de:13::a/64
DNS                :
ROUTER LINK-LAYER  :
MAC                : 00:50:79:66:68:04
LPORT              : 20046
RHOST:PORT         : 127.0.0.1:20047
MTU:               : 1500

VPCS>
```

Рисунок 16: Просмотр конфигураций

0.18 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

```
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
Welcome to the VyOS install program. This script
will walk you through the process of installing the
VyOS image to a local hard drive.
Would you like to continue? (Yes/No) [Yes]: yes
Probing drives: OK
The VyOS image will require a minimum 2000MB root.
Would you like me to try to partition a drive automatically
or would you rather partition it manually with parted?  If
you have already setup your partitions, you may skip this step

Partition (Auto/Parted/Skip) [Auto]: auto

I found the following drives on your system:
 sda    8589MB
 sdb    1MB


Install the image on? [sda]:sda

This will destroy all data on /dev/sda.
Continue? (Yes/No) [No]:
Ok then.  Exiting...
vyos@vyos:~$ reboot
```

0.19 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-02 - PuTTY

```
Login incorrect
vyos login: vyos
Password:
Linux vyos 4.19.195-amd64-vyos #1 SMP Sat Jun 19 08:48:00 UTC 2021 x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-kazhavoronkov-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-kazhavoronkov-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N]
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y
[  230.539300] reboot: Restarting system
```

0.20 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

msk-kazhavoronkov-gw-02 - PuTTY

```
Login incorrect
vyos login: vyos
Password:
Linux vyos 4.19.195-amd64-vyos #1 SMP Sat Jun 19 08:48:00 UTC 2021 x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-kazhavoronkov-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-kazhavoronkov-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N]
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] y
[  230.539300] reboot: Restarting system
```

0.21 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

```
VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.412 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=3.515 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=1.009 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.485 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=1.109 ms

VPCS> ping 2001:db8:c0de:12::a

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=5.610 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=1.021 ms
2001:db8:c0de:12::a icmp6_seq=3 ttl=62 time=1.096 ms
2001:db8:c0de:12::a icmp6_seq=4 ttl=62 time=0.795 ms
2001:db8:c0de:12::a icmp6_seq=5 ttl=62 time=2.308 ms
```

Рисунок 20: Проверка подключения с PC3

0.22 Настройка двойного стека адресации IPv4 и IPv6 в локальной сети

```
VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.412 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=3.515 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=1.009 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.485 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=1.109 ms

VPCS> ping 2001:db8:c0de:12::a

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=5.610 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=1.021 ms
2001:db8:c0de:12::a icmp6_seq=3 ttl=62 time=1.096 ms
2001:db8:c0de:12::a icmp6_seq=4 ttl=62 time=0.795 ms
2001:db8:c0de:12::a icmp6_seq=5 ttl=62 time=2.308 ms
```

Рисунок 21: Проверка подключения с PC4

1. Вывод

1.1 Вывод

В ходе выполнения лабораторной работы мы изучили принципы распределения и настройки адресного пространства на устройствах сети.